Gruhesh Sri Sai Karthik Kurra

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Imperial MSc (AI and ML) student working on language models, available from autumn 2027: first-author NLP paper in IEEE Xplore, two first-author papers accepted at IEEE NEPCON 2026 (diverse generation from a frozen transformer; deepfake detection), and a data-leakage audit that changed a paper's main result. Seeking research engineer roles and internships in LLMs.

Education

Imperial College London, United Kingdom Sep 2026 - Sep 2027 (expected)
MSc Computing (Artificial Intelligence and Machine Learning), Department of Computing | Enrolled

KL University, Hyderabad, India Aug 2021 - May 2025
B.Tech Computer Science and Engineering | CGPA 9.72 / 10

iHub-Data, IIIT Hyderabad, India May 2024 - Oct 2024
Six-month machine learning programme: classifiers, text representation, optimisation, clustering, PCA, probabilistic models, deep learning

Selected Publications

Morphology-Aware Nested Character Embeddings for Word Representation (MANCE)
Gruhesh Sri Sai Karthik Kurra, Chandanasree Moparthi, Prathipati Manish Chowdary, Pavan Kumar Pagadala
IEEE ICCCMLA 2025, published in IEEE Xplore
ieeexplore.ieee.org/document/11580466 | Code: github.com/GruheshKurra/MANCE-NLP

- Represented each word as an ordered sequence of character embeddings, so morphological variants share part of their representation, with a vocabulary of about 100 characters
- Re-implemented it as an open, seeded PyTorch pipeline (character Transformer word encoder, 3 seeds): 63.0 F1 on GermEval 2014 German NER against 48.2 for word embeddings with the same encoder, on par on AG News (91.0%) and DBpedia (98.5%); ablations showed encoding characters in order matters (averaging them cost 5.6 points)

Prism Tuning: Repulsion-Trained Seed Embeddings in a Frozen Transformer for Non-Redundant Generation
Gruhesh Sri Sai Karthik Kurra, Pavan Kumar Pagadala, Malathy Batumalay, Sushma Reddy Koduru
Accepted, IEEE NEPCON 2026 (presentation December 2026)

- Proposed training only K seed vectors, added to the input embeddings of a frozen transformer, with a repulsion loss, so one prompt yields K different outputs in one batched pass with no filtering at inference
- Made the loss trainable through a hidden-state surrogate at generated positions, whitened representations to counter anisotropy, and split each seed into shared style and repelled content parts (architecture and formulation only)
- Follow-up project, Prism-MCL (Oct 2026): replaced the loss with multiple-choice learning and tested prefix and per-layer attention seeds with LoRA on Qwen2.5 0.5B and 3B (RunPod); pre-registered runs on 42 held-out categories found no significant gain over tuned min-p

Dual-Stream Artifact Detection with Iterative Evidence Refinement for Frame-Level Deepfake Recognition
Gruhesh Sri Sai Karthik Kurra, Pavan Kumar Pagadala, Malathy Batumalay, Veda Boddapati
Accepted, IEEE NEPCON 2026 (presentation December 2026)
Frame corpus: kaggle.com/datasets/gruheshkurra/radar-deepfake-frames

- Audited the data split: 97.3% of WildDeepfake test frames shared source videos with training; reported the video-disjoint FaceForensics++ and Celeb-DF test subset (5,680 frames) as the main result: 96.3% AUC, and 87.1% on Celeb-DF
- Built a ViT-Small detector with Sobel-edge and three-band FFT evidence streams, an orthogonality penalty and iterative cross-attention fusion, trained on 227,504 frames from FaceForensics++, Celeb-DF v2 and WildDeepfake

BIE: Bit-Index Encoding of Sparse Neural Network Weights
Gruhesh Sri Sai Karthik Kurra
Preprint, Zenodo, version 2, October 2026
doi.org/10.5281/zenodo.17217217 | Code: github.com/GruheshKurra/bit-index-encoding-research-

- Designed a storage format for pruned weights (Golomb–Rice coded positions, 8-bit codes stored as bitplanes, row-parallel kernel): $44.5\times$ smaller than 32-bit weights at 95% sparsity with 0.6% error, against $25.7\times$ for CSR at the same error; perplexity of a pretrained 124M-parameter language model stays at the int8 level (30.0 vs 29.9)

Research Experience

Research Intern | IIIT Hyderabad Dec 2024 - Jun 2025
Hyderabad, India

- Designed a document translation system that keeps the original layout, using Vision Transformer layout detection, character-metric font detection and K-means colour extraction
- Built the optical character recognition pipeline with FastAPI and Tesseract for English, Hindi, Telugu and German, keeping source fonts, colours and page structure in the output
- Demonstrated the system live to faculty and industry visitors at the IIIT Hyderabad research expo, May 2025 (demo: youtu.be/rcvSuBcBjyg)

Industry Experience

AI Integration Engineer | InfoAjax Consulting Oct 2025 - Nov 2025
Remote, Hyderabad, India

- Built a working proof of concept of a browser agent that acts on screenshots (click, type) using models on Azure AI Foundry, with a React, FastAPI and WebSocket console to monitor agents live

Software Engineer (Freelance, WSO2 project) | InfoAjax Consulting Oct 2024 - Jun 2025
Remote, Hyderabad, India

- Built REST APIs for PLDT, a Philippine telecoms company, in WSO2 Micro Integrator, linking its service-desk, billing and inventory systems, and supported them in production for four months under service-level agreements

Technical Skills

Programming: Python, SQL, Java, C, Bash
NLP and ML: PyTorch, NumPy, Hugging Face Transformers, scikit-learn, pandas; Transformer, CNN and BiLSTM encoders, character-level representations, Vision Transformers
Research practice: experiment design, data-leakage audits, public dataset release, IEEE manuscripts
Engineering: FastAPI, REST, WebSocket, Docker, Git, GitLab, Linux, Azure AI Foundry

Awards and Leadership

1st Place, Global Challenge Lab, Imperial College London Jul 2026

- Led the AI and ML work in a five-person team (£1,500 prize, over 1,000 students) for OrbitOps, a reliability layer for AI on satellites: pre-launch fault injection, in-orbit known-answer probes and rollback

Technical Lead, Student Activity Council, KL University 2023 - 2025

- Mentored over 20 junior students in machine learning and web development, ran workshops for over 100 students; SAC Momentum Award